Pricing Capability: A Framework for Valuing Proprietary Data Assets in the Age of AI

Capability Alpha, Price-Implied Uplift, and the Spirit Aviation Data Auction

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Abstract

Proprietary enterprise data has become an input to model training, post-training, and evaluation, creating a valuation problem that spans intangible-asset finance and machine-learning data valuation. Google LLC's selected $10 million bid for Spirit Aviation Holdings' deidentified operational data and software provides a rare observable transaction in which this problem can be studied. Financial methods can value attributable cash flows, and machine-learning methods can estimate the contribution of data to model performance. The missing measurement is the link between a dataset's causal contribution to economically relevant model or system performance and its buyer-specific economic value.

We define *Capability Alpha* as the causal improvement in economically relevant model or system performance attributable to a proprietary dataset under controlled evaluation. The measured gain is translated into buyer-specific use value through economic exposure, the relationship between capability improvement and commercial outcomes, the expected duration of the gain, usable rights, and integration costs. We also define *Price-Implied Uplift* (PIU), an application of reverse discounted-cash-flow analysis that recovers the persistent attributable commercial improvement required to justify an observed transaction price.

Applied to the Spirit transaction, a scenario in which products or workflows representing 5% of annual Google Cloud revenue form the relevant economic base implies a PIU of approximately 0.236% over three years at the selected $10 million price. Separate conditional use-value scenarios produce buyer-specific values of approximately $50 million to $200 million. Across three additional recurring licenses involving public-company buyers, a standardized 5% scope produces PIU thresholds of approximately 0.58%–1.42%; Spirit's case-specific 0.236% threshold is numerically lower under its stated assumptions. We further consider buyer concentration, exclusivity, and preemption as sources of divergence between transaction price and buyer-specific value. Spirit's Capability Alpha has not been measured empirically. The quantitative results are conditional valuation scenarios and do not constitute an appraisal.

Keywords: data valuation; AI training data; model evaluation; intangible assets; reverse DCF; industrial organization.

1 Introduction

Recent data partnerships and post-training studies indicate that proprietary enterprise records are becoming economically relevant inputs to model development. These assets include expert demonstrations, operational workflows, internal software repositories, communications, transaction records, and linked records of decisions and outcomes. Work on the economics of AI training data documents a shift from public web corpora toward proprietary sources and identifies context-dependent value and contribution to production as open measurement problems [1]. Empirical work from Mercor and Harvey further indicates that expert and operational data can materially affect model performance in professional settings [2, 3].

An active market is already visible in publicly reported AI-data and content agreements. Reddit's Google license was reported at approximately \$60 million per year; News Corp's OpenAI partnership at more than \$250 million over five years; the New York Times' Amazon agreement at approximately \$20 million–\$25 million per year; and News Corp's Meta agreement at up to \$50 million per year [4, 5, 6, 7]. The rights, asset types, and commercial uses differ across these contracts, but their scale shows that AI-related data access is already being priced at economically material levels.

The valuation problem sits between two established literatures. Intangible-asset finance provides cost, market, and income approaches for valuing economic benefits attributable to an asset [8, 9, 10, 11]. Machine-learning data valuation provides methods for estimating how observations or datasets contribute to model performance [12, 13, 14, 15]. Evaluations such as SWE-Lancer and GDPval increasingly measure model performance on economically valuable, real-world tasks [16, 17]. What remains underdeveloped is a method for taking a causal performance contribution measured on economically relevant tasks and translating it into buyer-specific cash flows.

This paper introduces *Capability Alpha* for that purpose. Capability Alpha is the causal improvement in economically relevant model or system performance attributable to a proprietary dataset under a controlled evaluation. Buyer-specific use value is then determined by the economic activity exposed to the affected task, the mapping from performance improvement to economic outcome, the duration of the gain, usable rights, and acquisition and integration costs. Statistical and attribution uncertainty are reported with the Capability Alpha estimate, for example through confidence intervals or scenario ranges. The economic equation therefore does not include a separate uncertainty multiplier.

The framework is applied to Spirit Aviation Holdings' proposed sale of deidentified data and software assets to Google LLC for \$10 million [18]. The public filing describes a large operational corpus spanning communications, IT records, source code, pricing, revenue, airline operations, maintenance, finance, legal, and employee records. The transaction provides observable bids, a detailed asset schedule, and contractual terms, allowing the assumptions implied by the selected price to be stated explicitly.

Two quantitative objects are used in the case study. A forward use-value analysis asks what the asset could be worth to a particular buyer under stated assumptions about economic impact and value capture. A reverse calculation, *Price-Implied Uplift* (PIU), starts from the observed price and solves for the persistent attributable commercial improvement required for the transaction to break even. Under an illustrative scenario in which products or workflows representing 5% of annual Google Cloud revenue form the relevant economic base, the selected \$10 million price implies a PIU of approximately 0.236% over three years. Separate conditional use-value scenarios produce values of approximately \$50 million to \$200 million.

Market structure is treated separately from direct use value. The set of buyers able to monetize a proprietary corpus can be small because model development, post-training infrastructure, compute, distribution, and downstream economic exposure are complementary inputs [19, 20]. Restrictive rights may also create preemption value when rival access has a measurable competitive effect.

These mechanisms can cause transaction price to diverge from buyer-specific economic value.

2 Related literature and positioning

The framework draws on five literatures: intangible-asset valuation, machine-learning data valuation, evaluation on economically valuable tasks, the economics of AI production, and industrial organization. This section identifies the established components and isolates the remaining measurement problem.

2.1 Intangible assets and data valuation

The valuation literature already provides several ways to value intangible assets. Cost approaches estimate the resources required to create or replace an asset. Market approaches use comparable transactions. Income approaches discount the incremental cash flows attributable to an asset. International Valuation Standards also distinguish Market Value from Investment Value or Worth, which is specific to an owner or prospective owner, and Synergistic Value, which can arise from combinations of assets or interests [10, 11].

Data creates additional measurement problems. Coyle and Manley survey empirical data-valuation methods and find no consensus method that works across data types and purposes [8]. The OECD similarly emphasizes that data value depends on use, access, governance, and rights [9]. Mohan's review of enterprise data valuation documents the continuing gap between the recognized economic importance of data and the practical methods available to firms [21]. Recent macroeconomic work also treats data as an asset with a depreciating productive value, reinforcing the need to model economic life explicitly [22].

Recent work on the economics of AI training data makes the same measurement problem explicit. Oderinwale and Kazlauskas characterize data as a distinct input to AI production and identify context-dependent value and estimation of data's contribution to production as central open problems [1]. For proprietary training and evaluation data, the income approach therefore provides a natural economic endpoint. It still requires a credible estimate of the causal economic benefit attributable to the dataset.

2.2 Machine-learning data valuation

Machine-learning research approaches the same problem from the technical side. Data Shapley assigns value to training observations according to their marginal contribution to model performance [12]. Jia et al. develop more efficient Shapley-based estimation methods for practical data valuation [13]. Privacy-preserving work extends this logic to marketplace settings where data owners must be compensated without exposing the underlying data [15].

More recent work moves toward utility-based pricing for LLM training data. Xu et al. combine token-level information measures with proxy-model training gain, influence methods, and Data Shapley, explicitly arguing that row count and token count are weak proxies for training utility [14]. This work moves closer to empirical utility measurement. The remaining gap for asset valuation is economic translation. A measured contribution to benchmark performance does not itself tell a buyer whether a corpus is worth thousands, millions, or hundreds of millions of dollars.

Capability Alpha is proposed as an intermediate object between these two layers. It measures the causal contribution of a corpus to economically relevant model or system performance; the valuation layer then translates that contribution into buyer-specific economic value.

2.3 Evaluation on economically valuable tasks and expert data

The evaluation literature is increasingly able to measure model performance on economically valuable, real-world tasks. SWE-Lancer contains more than 1,400 freelance software-engineering tasks associated with $1 million in actual payouts [16]. GDPval evaluates 1,320 tasks across 44 occupations selected for their economic significance and uses work products created by experienced professionals [17]. GDPval explicitly frames this direction as the measurement of economically relevant capabilities. These benchmarks provide task-level evidence that is closer to production work than conventional academic benchmarks.

Evidence from post-training provides a second empirical input: proprietary expert data can produce measurable changes in model performance. Mercor reports that 874 expert-created tasks materially improved GLM 4.6 on APEX-Agents, with Pass@1 and mean scores close to doubling and corporate-law Pass@1 tripling [2]. Harvey's Tenet work shows gains from a stack combining public, synthetic, and human-expert data with post-training and harness changes [3]. Harvey also reports that harness changes alone can materially move benchmark performance, which is important for attribution.

These results motivate a controlled design. A before-and-after comparison of a model or system configuration is insufficient when data, harness, tools, compute, rewards, and model checkpoints move together. Valuation requires the marginal effect of the proprietary corpus to be identified separately.

2.4 Economics of AI and buyer concentration

The same capability gain can have different economic value across buyers. Demirer et al. document substantial variation in demand for model capability across applications using OpenRouter and Microsoft Azure usage data [23]. Gans argues that data markets, compute, model openness, and other bottlenecks interact to determine AI market power [19]. Korinek and Vipra emphasize economies of scale and scope in foundation-model development and the concentration of key inputs such as compute, data, and talent [20].

These observations matter for data transactions. The set of firms that can technically receive a dataset can be much larger than the set that can economically monetize it at scale. Buyer-specific value can therefore vary by orders of magnitude. Thin buyer markets can also make observed prices sensitive to bargaining power, auction design, and the value of the next credible bidder.

2.5 Exclusivity, foreclosure, and preemption

Exclusive or restrictive rights add a second strategic channel. Competition economics has long studied cases where control of an input changes rivals' costs or access. OECD work on digital mergers discusses data as a potentially important competitive input and treats input foreclosure as a relevant theory of harm in some settings [24]. More recent OECD work on AI downstream markets highlights data access, vertical integration, model restrictiveness, and control of enabling inputs as factors that can shape competition [25].

Preemption also has an established place in models of strategic investment. Weeds studies R&D investment under uncertainty where competition for a winner-takes-all asset changes investment timing through preemption incentives [26]. For data, the analogous mechanism is narrower: a buyer may receive economic value when exclusive rights prevent a close rival from acquiring a capability-relevant input.

The contribution here is to place this strategic component alongside direct use value while keeping the two separate. In the Spirit case, the quantitative range in this paper excludes a preemption premium because the public record does not establish such a motive or quantify the effect of rival access.

2.6 Research gap

The literature therefore provides most of the component parts. Finance can value attributable economic benefits. Machine-learning methods can estimate data contribution to model performance. Evaluations on economically valuable tasks provide increasingly realistic measures of model performance. Industrial organization explains why buyer concentration and exclusivity can affect price. The unresolved step is estimating the causal contribution of a proprietary dataset to economically relevant model or system performance, translating that contribution into buyer-specific cash flows, and comparing the result with an observed or proposed transaction price. The remainder of the paper formalizes that step.

3 Framework

3.1 Core concepts

Three concepts organize the measurement layer of the framework.

Frontier Deficit. For a task family, define a reference level of performance and compare it with current frontier performance. The gap is Frontier Deficit. A large gap indicates that valuable capability is still missing; the dataset's ability to close that gap must be measured separately.

Capability Alpha. For dataset D and task family t, Capability Alpha, written $CA_{D,t}$, is the causal improvement in economically relevant model or system performance attributable to D under controlled evaluation. Operationally, it is the performance difference attributable to adding the proprietary dataset while the remaining experimental stack is held fixed. Statistical and attribution uncertainty are reported with the Capability Alpha estimate, for example through confidence intervals or scenario ranges.

Verifier Density and Outcome Linkage. Some datasets contain linked cases where one can reconstruct context, action, and outcome. Others are mostly disconnected text. Verifier Density describes how much of a corpus supports objective or structured judgment of success. Outcome Linkage describes how reliably records can be connected across workflow stages. These properties matter for post-training because useful trajectories can supply demonstrations, evaluation environments, and reward signals.

Figure 1 summarizes the valuation pipeline.

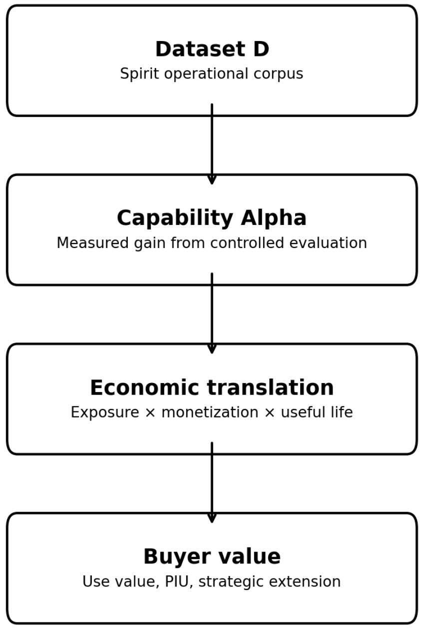

Figure 1: Core valuation pipeline.

3.2 Direct use value

The direct use value of a dataset to buyer b is the present value of the economic gains attributable to Capability Alpha, adjusted for rights and usability, minus acquisition-adjacent costs:

$$V_b^{use}(D) = \rho_D \sum_h \frac{1}{(1+r)^h} \sum_t (E_{b,t,h} \lambda_{b,t} CA_{D,t} d_{D,t,h}) - C_D. \tag{1}$$

Here t indexes economically relevant task families and h indexes future years over the modeled economic horizon. Each term has a simple interpretation. $E_{b,t,h}$ is the annual economic exposure: the amount of business that sits under task family t in year h. $\lambda_{b,t}$ is the monetization coefficient: how much a unit of capability improvement turns into revenue, cost savings, throughput, or another economic outcome. $CA_{D,t}$ is the controlled performance gain attributable to dataset D. $d_{D,t,h}$ is persistence: how much of the current data advantage remains useful in future year h.

The term r is the discount rate. It converts future cash flows into present value. Inflation can influence the discount rate, and the rate also reflects risk and opportunity cost. A future dollar of profit has a lower present value than a dollar received immediately. This is distinct from $d_{D,t,h}$, which captures whether the data advantage itself decays as models improve, substitutes appear, or business processes change [11, 22].

The term ρ_D is the rights and usability adjustment. It captures how much technical and economic utility survives privacy constraints, deidentification requirements, license restrictions, and similar limits. C_D collects acquisition-adjacent costs such as legal work, deidentification, engineering, cleaning, training, and integration.

A simpler reading of Equation 1 is:

Direct use value equals the economic activity exposed to the task, multiplied by the performance gain attributable to the dataset and the rate at which that gain translates into economic outcomes, with adjustments for persistence, present value, usable rights, and costs.

3.3 Price-Implied Uplift

When transaction price P is observable and Capability Alpha has not yet been measured, the valuation can be inverted. Suppose the buyer has a relevant revenue base R_{scope}, operating margin m, useful life T, discount rate r, rights adjustment ρ, and costs C. In this simplified specialization, ρ is the scalar counterpart of ρ_D in Equation 1, and C is the corresponding acquisition-adjacent cost term. The persistent attributable commercial uplift needed to justify price P is:

$$\text{PIU} = \frac{P + C}{R_{\text{scope}} m A(T, r) \rho}, \tag{2}$$

where $A(T, r) = \sum_{h=1}^{T} (1+r)^{-h}$ is the discounted annuity factor. This is an application of reverse discounted-cash-flow logic to a data transaction [27].

PIU is an economic threshold. If PIU equals 0.236%, the claim is that the data must improve the economics of the modeled business slice by 0.236% on a persistent basis for the observed purchase price to break even under the stated assumptions. The realized effect remains an empirical quantity until the dataset is evaluated.

3.4 Strategic extension

Equation 1 measures direct use value. A strategic extension can be added explicitly when the relevant effect is separately identified:

$$V_b^{total}(D) = V_b^{use}(D) + S_b(D). \tag{3}$$

Here $V_b^{total}(D)$ is buyer b's total value for dataset D, and $S_b(D)$ is the strategic component outside direct use value. The strategic component can include future option value and, when the rights package is sufficiently restrictive, preemption value:

$$S_b(D) = V_b^{preemption}(D) + V_b^{option}(D) + \ldots \tag{4}$$

Here $V_b^{option}(D)$ denotes the present value of separately identified future uses of D that are not already included in direct use value. For buyer b, preemption value can be written as:

$$V_b^{preemption}(D) = \Pi_b(\text{rivals without } D) - \Pi_b(\text{rivals with } D). \tag{5}$$

Here $\Pi_b(\cdot)$ denotes the present value of buyer b's expected profit under the indicated rival-access state. This term should be estimated only when there is evidence that rights restrict a rival's access and that the rival would receive a meaningful capability gain from the data. In the Spirit calculations below, $S_b(D)$ is left outside the quantitative range.

4 Measuring Capability Alpha

The empirical validity of the valuation depends on a controlled estimate of the dataset's contribution to model or system performance.

The dataset should first be mapped into economically meaningful task families. Depending on the corpus, these may include pricing, disruption handling, software maintenance, forecasting, fraud investigation, contract review, reconciliation, or customer support. Each task family needs a held-out benchmark and a calibrated verifier, using objective outcomes where available and expert rubrics where necessary. Frontier-model baselines should be evaluated first to estimate the remaining capability gap.

The experimental design should hold the model family, compute budget, harness, and evaluation setup fixed while varying access to the proprietary dataset. A factorial design can separate the effects of harness changes, public data, synthetic data, proprietary data, tools, and reward design. Harvey's results illustrate why this matters: harness changes alone can move evaluation performance [3].

The evaluation should estimate Capability Alpha with uncertainty, then repeat the treatment at increasing data fractions to measure saturation. Repeating the experiment across model families tests whether the estimate transfers beyond a single checkpoint or architecture. Mercor's reported post-training experiment provides an example of the type of evidence this framework needs: held-out worlds, controlled training, and explicit ablations around a small expert dataset [2].

The final step is economic translation. Production telemetry or a transparent scenario model is needed to estimate the relevant monetization coefficient $\lambda_{b,t}$, the relationship between measured performance improvement and economic outcome for buyer b and task family t. This can use completion rate, human-review time, cost per accepted outcome, throughput, error cost, conversion, retention, or revenue. When production evidence is unavailable, the paper should report scenario ranges and identify the assumptions that drive them.

5 Spirit Aviation case study

5.1 Transaction and asset

Spirit Aviation Holdings selected Google LLC as the winning bidder for a set of deidentified data and software assets at a purchase price of \$10 million. Mercor was the \$7.5 million alternate bidder. Micro1 later announced a \$12.5 million offer outside the initial auction process [18, 28]. Court approval remained pending as of the public record used in this paper, with employee-privacy objections affecting the approval process [29].

The asset schedule describes a broad operating corpus. It includes approximately 100 million emails, 500 million Microsoft Teams items, 17.1 million OneDrive items, 20.6 million SharePoint items, 667,563 IT tickets, 516 code repositories containing roughly 30 million lines of code, 372,585 commits, 3.53 billion booking-curve observations, 7.25 billion competitor-flight observations, 7.51 billion revenue transactions, and extensive airline operations, maintenance, finance, legal, and employee records [18].

The agreement requires deidentification while preserving referential integrity across the data structure [18]. That provision is economically relevant because stable relationships across records may preserve workflow trajectories. A pricing state could be linked to a decision and booking outcome. An IT ticket could be linked to a code change and deployment result. An irregular operation could be linked to a reaccommodation decision and later operational outcome. The extent of these linkages is unknown without direct inspection.

The transferred assets are a bounded subset of Spirit's data estate. The filing excludes major consumer datasets, including customer profiles and Free Spirit member records [18]. The paper therefore avoids describing the transaction as a sale of all Spirit data.

5.2 Why the asset could matter to Google

Google LLC is the buyer. Google Cloud is used as an illustrative commercial surface because public segment revenue and operating margin are available [30]. The calculation uses those disclosed segment economics as a transparent scenario base; the public record does not identify Google Cloud as the exclusive deployment surface for the Spirit assets.

Google Cloud already supports airline optimization and decision-support use cases. Its SWISS case study describes operational decision support using crew, passenger, rotation, and technical data, with reported savings from rotation optimization [31]. Its FLYR case study describes machine-learning revenue management for airlines and reports material revenue effects in customer deployments [32]. These examples establish that airline operations and revenue-management tasks can have measurable commercial consequences within Google Cloud deployments. They provide no estimate of Spirit's contribution.

6 Spirit valuation scenarios

6.1 Observed bids

The observed process gives three price signals: Mercor at \$7.5 million, Google selected at \$10 million, and micro1's later \$12.5 million indication [18, 28]. The micro1 figure should be treated as stated willingness to pay because it was outside the initial auction process and had not become a completed transaction in the record used here.

Figure 2 places these observed figures next to the use-side scenario range developed below. The two objects serve different purposes. The points are market evidence. The interval is a conditional buyer-specific valuation scenario.

6.2 A use-side scenario range

The headline range starts from global airline revenue of roughly \$1.165 trillion in 2026 [33]. Suppose a capability influenced by Spirit-like operational data contributes to a 1% improvement in airline economics. That corresponds to \$11.65 billion of annual economic improvement across the industry. The 1% figure is an illustrative scenario input; Spirit's causal effect remains unmeasured.

Now suppose Google captures 0.5% to 2.0% of that created value through Google Cloud and related model-enabled products or services. That translates into roughly \$58.25 million to \$233 million of annual incremental Google revenue. Alphabet reported Q2 2026 Google Cloud revenue of \$24.768 billion and operating income of \$8.814 billion, an operating margin of about 35.6% [30]. Applying that margin gives approximately \$20.7 million to \$82.9 million of annual operating profit.

Using a three-year economic life and a 12% discount rate gives a discounted annuity factor of approximately 2.402. The resulting present-value range is roughly \$49.8 million to \$199.2 million, rounded to \$50 million to \$200 million.

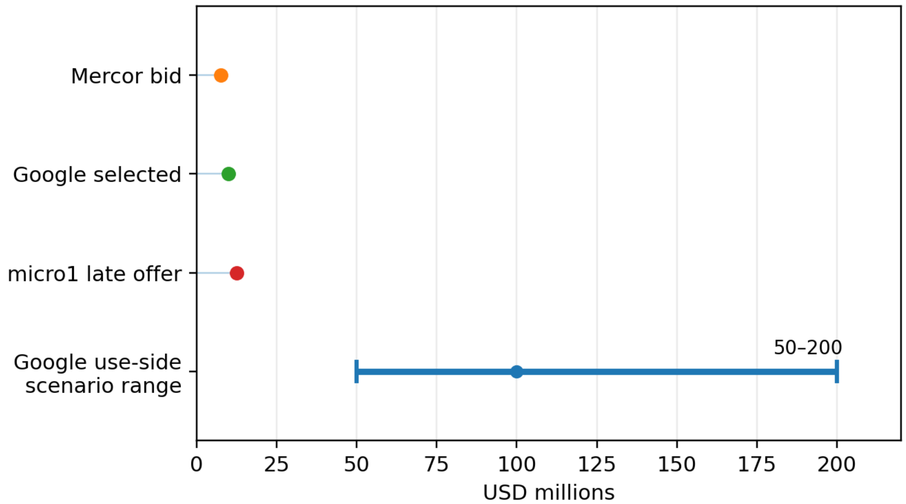

Figure 2: Observed bids and one buyer-specific use-side scenario range.

This range is a conditional buyer-specific use-value scenario, with no claim to fair value. The paper has no empirical basis for selecting the 1% industry improvement or 0.5% to 2.0% capture assumptions as central forecasts; the calculation shows which assumptions generate values of this order.

6.3 Price-Implied Uplift for Spirit

PIU starts from the observed \$10 million price. Google Cloud's Q2 2026 revenue annualizes to approximately \$99.072 billion [30]. The paper then defines a *relevant revenue scope*: the share of annual Google Cloud revenue associated with products or workflows that could plausibly experience an economic effect from capabilities affected by the dataset.

For the 5% scenario:

$$R_{\text{scope}} = 0.05 \times \$99.072\text{B} = \$4.954\text{B}.$$

The 5% parameter sets only the size of the modeled economic base. Under this scenario, products or workflows representing \$4.954 billion of annual Cloud revenue are treated as within scope for a possible effect from the capability; no claim is made about aviation's share of Cloud revenue or a 5% Spirit-driven revenue increase.

Applying the 35.6% operating margin gives approximately \$1.763 billion of annual operating profit within scope. Discounting three years at 12% gives a present-value profit base of about \$4.23 billion. A \$10 million purchase price then implies:

$$\text{PIU} = \frac{\$10\text{M}}{\$4.23\text{B}} \approx 0.236\%.$$

Table 1 and Figure 3 show the sensitivity to the scope assumption.

Relevant Cloud revenue scope	Annual revenue base	PIU for \$10M price
1%	\$0.991B	1.181%
2.5%	\$2.477B	0.472%
5%	\$4.954B	0.236%
10%	\$9.907B	0.118%

Table 1: Price-Implied Uplift under alternative revenue-scope assumptions.

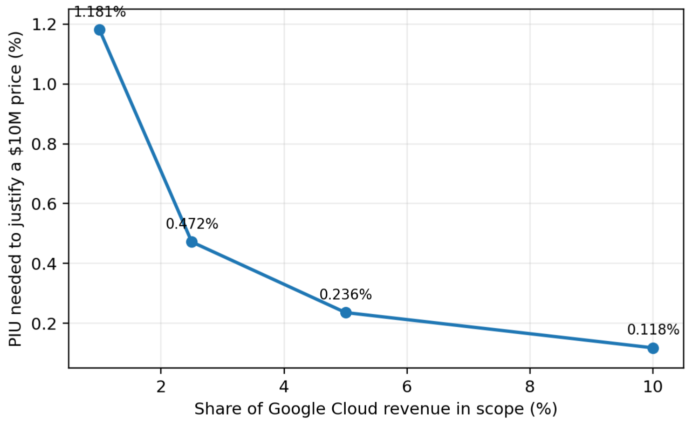

Figure 3: PIU declines as the modeled revenue scope grows.

Under the 5% scope assumption, a persistent attributable improvement of approximately 0.236% in the economics of the scoped Google Cloud revenue base is sufficient for a \$10 million purchase to break even over the modeled three-year horizon. The empirical question is whether measured Capability Alpha, after translation into production economics, exceeds that threshold.

Cross-transaction reference range. Table 2 places Spirit beside three recurring AI-data/content licenses for which public-company financials permit a standardized calculation. For the recurring licenses, the economic scope is fixed at 5% of the buyer's pre-deal consolidated operating-income base. Those standardized thresholds range from approximately 0.58% to 1.42%. Spirit uses the separate Google Cloud scenario developed above and yields 0.236%. The contracts differ in duration, rights, asset type, and economic base. The panel should be read as a reference range; direct valuation comparability is limited.

Transaction	Reported consideration	PIU, stated 5% scope
Spirit → Google	$10M selected package [18]	0.236% (case-specific)
NYT → Amazon	$20M–$25M/year reported [6]	0.583%–0.729%
News Corp → Meta	Up to $50M/year reported [7]	≤1.201%
Reddit → Google	∼$60M/year reported [4]	1.424%

Table 2: Cross-transaction PIU reference range. Recurring-license cases use a standardized 5% share of pre-deal consolidated operating income; Spirit uses the case-specific Google Cloud assumptions in Section 6.3. Full derivations appear in Appendix A.

Under these respective assumptions, Spirit's threshold is numerically below the 0.58%–1.42% standardized range for the recurring-license cases. The comparison cannot establish underpricing. Under the stated assumptions, the selected Spirit price has the lowest break-even hurdle in this small public panel. A relatively small persistent economic contribution from the acquired package could therefore justify the observed price for Google under the modeled scope.

The Amazon case also shows how strongly the economic denominator matters. Using 5% of AWS operating income in place of 5% of consolidated Amazon operating income raises the reported-fee midpoint threshold from approximately 0.66% to 1.13%. This sensitivity supports the framework's buyer- and use-specific exposure term. The News Corp–OpenAI agreement is excluded from the point comparison because the public record provides no contemporaneous positive operating-margin denominator and reported consideration mixes cash with credits. Appendix A reports the full derivations, scope sensitivities, and evidence hierarchy.

7 Market structure and strategic value

The Spirit process also illustrates why buyer-specific value and transaction price can diverge. A proprietary corpus can be technically transferable to many firms while only a small number possess the models, compute, training infrastructure, distribution, and commercial exposure needed to monetize it at scale [19, 20]. In a thin buyer market, the winning buyer's value can be far above the price needed to beat the next credible bidder.

Exclusivity changes the analysis further. A restrictive rights package can create preemption value when a close rival would gain meaningful capability from the same data and cannot obtain an equivalent substitute [24, 25, 26]. The strategic extension in Equation 3 captures that possibility. The Spirit scenario range excludes it because the public record does not provide evidence sufficient to estimate a competitive-denial component.

Technology can change both sides of the market over time. Stronger open models, cheaper compute, and more accessible post-training infrastructure can expand the set of economically capable buyers. The same technological progress can reduce the use value of a fixed corpus if frontier-model performance closes the relevant capability gap or substitute data and synthetic environments improve. Asset value and seller bargaining power can therefore move in different directions.

8 Limitations and falsification

The main limitation is that Spirit's actual Capability Alpha remains unmeasured. The $50 million to $200 million figure is a scenario range, and PIU is a threshold calculation conditional on scope, margin, duration, rights, and cost assumptions.

Attribution is a second limitation. Model gains can come from data, model family, compute, prompting, tools, retrieval, reward design, or harness engineering. A controlled evaluation is needed to isolate the proprietary corpus. Benchmark gains can also fail to transfer into production economics.

Rights and usability create a third source of uncertainty. A technically useful corpus can lose economic value through privacy restrictions, deidentification burdens, broken linkage, contractual limits, or high integration costs. Economic life is also uncertain because frontier progress and substitute data can erode scarcity [11, 22].

The Spirit undervaluation thesis would weaken if the deidentified asset turns out to have low practical usability, weak Outcome Linkage, low Verifier Density, little transfer to economically relevant tasks, or measured Capability Alpha below the PIU threshold under a reasonable scope assumption. Evidence in the other direction would strengthen the buyer-specific value case.

9 Conclusion

This paper develops a framework for valuing proprietary data from its causal contribution to economically relevant model or system performance. Capability Alpha provides the technical input: a controlled estimate of the performance gain attributable to the dataset. Buyer-specific use value then depends on economic exposure, the translation from performance to commercial outcomes, persistence, usable rights, and costs. PIU provides the inverse calculation by recovering the persistent attributable commercial improvement required to justify an observed transaction price. Strategic effects such as preemption are treated as a separate extension and should be included only when the rights and competitive evidence support them.

The Spirit Aviation transaction provides a public case in which the framework can be applied to observable bids and a disclosed asset schedule. Under the conditional use-value assumptions developed here, buyer-specific values of approximately $50 million to $200 million are economically coherent. Under a scenario in which products or workflows representing 5% of annual Google Cloud revenue form the relevant economic base, the selected $10 million price implies a PIU of approximately 0.236% over three years. Both results are conditional on explicit assumptions that can be tested or revised and carry no appraisal claim.

Across three additional recurring transactions involving mature public-company buyers, the standardized 5% scope produces PIU thresholds of approximately 0.58%–1.42%. Spirit's case-specific 0.236% threshold is numerically lower under its stated assumptions. Contractual and economic differences limit direct comparison, but the panel provides a useful cross-transaction reference range for the break-even economics implied by observed AI-data consideration. Detailed calculations are reported in Appendix A.

Empirical validation requires a legally usable enterprise corpus with linked workflows and observable outcomes. A controlled study can construct economically relevant task families, estimate Capability Alpha with uncertainty, measure the translation from performance to production outcomes, and compare the resulting value with PIU and observed transaction prices. Repeated applications would allow the framework to be calibrated against realized model gains, buyer economics, and market outcomes.

A Public transaction calculations and source notes

A.1 Evidence hierarchy and recurring-license derivation

This appendix applies PIU to observed or reported AI-data/content transactions beyond Spirit. Evidence is separated into three layers: terms disclosed by a contracting party or securities filing; the existence and stated purpose of a partnership disclosed by the parties; and consideration reported by a financial-news source when price remains confidential. Press-reported prices are labeled as such throughout.

For a one-time purchase price P, Equation 2 gives

$$\mathrm{PIU} = \frac{P + C}{R_{\mathrm{scope}} m A(T,r)\rho}, \qquad A(T,r) = \sum_{h=1}^{T} \frac{1}{(1+r)^h}.$$

The symbols P, C, R_{scope}, m, T, r, and ρ retain the definitions from Section 3.3. For recurring-license cases, let F denote level annual consideration, R the buyer's annual revenue, and s the assumed share of that revenue within the modeled economic scope, so $R_{\mathrm{scope}} = sR$. For a level attributable annual benefit over the same horizon,

$$\mathrm{PIU}_{\mathrm{annual}} = \frac{F A(T,r) + C}{R_{\mathrm{scope}} m A(T,r)\rho}.$$

With $C = 0$, $\rho = 1$, and matched annual timing of fee and benefit, the annuity factor cancels:

$$\boxed{\mathrm{PIU}_{\mathrm{annual}} = \frac{F}{R_{\mathrm{scope}} m}} = \frac{F}{sRm}.$$

Let OI denote the buyer's annual operating income. When consolidated operating margin is used as the standardized proxy, $m = \mathrm{OI}/R$, and therefore

$$\boxed{\mathrm{PIU}_{\mathrm{annual}} = \frac{F}{s\,\mathrm{OI}}}.$$

Consolidated operating margin is a transparent cross-case baseline and may differ materially from the incremental margin of the affected AI product. The case sheets use the most recent completed fiscal year available before each transaction and report scope sensitivity at 1%, 2.5%, 5%, and 10% when public-company financials permit the calculation.

The literature supports treating these contracts as heterogeneous [1, 34, 35]. Public terms vary across pricing mechanism, rights, duration, and usage, and many bilateral details remain confidential. The exercise converts observed consideration into explicit buyer-side break-even hypotheses; universal "price per token" or "price per article" estimates fall outside its scope.

A.2 Reddit → Google, 2024

Transaction evidence. Reuters reported on February 22, 2024 that Reddit had licensed its content to Google for approximately \$60 million annually and that Google could use Reddit content to train AI models [4]. Reddit's S-1, filed the same day, independently disclosed January 2024 data-licensing arrangements with \$203.0 million of aggregate contract value, two- to three-year terms, and at least \$66.4 million expected to be recognized in 2024 [36]. The filing states that substantially all contract value associated with licensing revenue came from one partner and describes model training as a licensing use. Because the S-1 does not name Google or provide a Google-specific price, the \$60 million annual figure remains a Reuters-reported term.

Buyer baseline. Alphabet's FY2023 Form 10-K reports \$307.394 billion of revenue and \$84.293 billion of operating income [37]. With annual fee $F = \$60$ million and standardized exposure $s = 5\%$,

$$\mathrm{PIU}_{5\%} = \frac{\$60\mathrm{M}}{0.05 \times \$84.293\mathrm{B}} = 1.424\%.$$

Standardized Alphabet scope	Scoped FY2023 operating income	PIU at \$60M/year
1%	\$0.843B	7.118%
2.5%	\$2.107B	2.847%
5%	\$4.215B	1.424%
10%	\$8.429B	0.712%

Table 3: Reddit–Google recurring-license PIU sensitivity using Alphabet FY2023 consolidated operating income as a standardized proxy.

At the 5% standardized scope, a \$60 million annual fee corresponds to approximately 1.42% of persistent revenue-equivalent improvement. A narrower economic base raises the threshold and a broader base lowers it. Reddit also provides a continuously renewing platform corpus, unlike Spirit's bounded enterprise archive.

A.3 News Corp → OpenAI, 2024

Transaction evidence. News Corp and OpenAI announced a multi-year global partnership covering current and archived content from major News Corp publications, with content display in responses and product enhancement [38]. Reuters reported that the agreement could be worth more than \$250 million over five years [39]; the Wall Street Journal reported that consideration includes cash and OpenAI credits and that content may be used for queries and training [5]. The reported contract value therefore mixes forms of consideration.

No point estimate. OpenAI was private and did not publish audited product-level economics. Later 2024 reporting described approximately \$3.7 billion of expected 2024 revenue and an approximately \$5 billion loss [40]. A current consolidated operating-margin denominator would be negative and economically unsuitable for this calculation. A forward model is required.

Treating \$250 million over five years as a nominal annualized contract-value floor of \$50 million per year, ignoring the cash/credit mix and growth for illustration, gives

$$\mathrm{PIU}^{\mathrm{illustrative}} = \frac{\$50\mathrm{M}}{s \times \$3.7\mathrm{B} \times m^{+}},$$

where m^+ is an assumed positive contribution margin.

Revenue scope s	$m^+ = 25\%$	$m^+ = 50\%$	$m^+ = 75\%$
25%	21.62%	10.81%	7.21%
50%	10.81%	5.41%	3.60%
100%	5.41%	2.70%	1.80%

Table 4: Illustrative News Corp–OpenAI threshold sensitivity. Public information is insufficient for a transaction PIU point estimate because cash-equivalent consideration, contemporaneous positive contribution margin, and product exposure are undisclosed.

A defensible valuation would require a forward revenue path, product contribution margin, expected useful life, the cash-equivalent value of credits, and potentially a strategic or option-value term. The sensitivity makes those missing primitives explicit.

A.4 The New York Times → Amazon, 2025

Transaction evidence. Reuters reported that the New York Times Company entered its first generative-AI licensing agreement with Amazon in May 2025. The multi-year agreement covers The New York Times, NYT Cooking, and The Athletic, permits real-time summaries and excerpts in products such as Alexa, and permits training of Amazon's proprietary foundation models [41]. The Wall Street Journal reported annual consideration of approximately \$20 million–\$25 million and described the arrangement as cash-only [6].

Buyer baseline. Amazon's FY2024 Form 10-K reports consolidated net sales of \$637.959 billion and operating income of \$68.593 billion. AWS separately reported \$107.556 billion of sales and \$39.834 billion of operating income [42]. At the midpoint annual fee of \$22.5 million and 5% consolidated scope,

$$\mathrm{PIU}_{5\%,\,\mathrm{mid}} = \frac{\$22.5\mathrm{M}}{0.05 \times \$68.593\mathrm{B}} = 0.656\%.$$

The reported \$20 million–\$25 million range maps to 0.583%–0.729% under the same scope.

Standardized Amazon scope	Scoped FY2024 operating income	PIU at \$22.5M/year
1%	\$0.686B	3.280%
2.5%	\$1.715B	1.312%
5%	\$3.430B	0.656%
10%	\$6.859B	0.328%

Table 5: New York Times–Amazon recurring-license PIU sensitivity using the reported fee midpoint.

The business-unit choice illustrates model risk. Comparing the same midpoint fee with 5% of FY2024 AWS operating income gives

$$\frac{\$22.5\mathrm{M}}{0.05 \times \$39.834\mathrm{B}} = 1.130\%.$$

The agreement spans Alexa and foundation-model training, so buyer-side diligence would need to map economic exposure across those uses. The consolidated and AWS calculations are sensitivity anchors.

A.5 News Corp → Meta, 2026

Transaction evidence. Meta publicly announced News Corp as a content partner for Meta AI in March 2026, describing real-time content integration [43]. The Wall Street Journal reported that the multi-year agreement could pay News Corp up to \$50 million per year for at least three years, with rights covering current information and archives [7]. The public figure is an annual cap.

Buyer baseline. Meta's FY2025 Form 10-K reports revenue of \$200.966 billion and income from operations of \$83.276 billion [44]. At the reported annual cap and 5% standardized scope,

$$\mathrm{PIU}_{5\%} \leq \frac{\$50\mathrm{M}}{0.05 \times \$83.276\mathrm{B}} = 1.201\%.$$

If realized annual consideration is below \$50 million, the corresponding threshold is lower.

Standardized Meta scope	Scoped FY2025 operating income	PIU at \$50M/year cap
1%	\$0.833B	≤6.004%
2.5%	\$2.082B	≤2.402%
5%	\$4.164B	≤1.201%
10%	\$8.328B	≤0.600%

Table 6: News Corp–Meta upper-bound recurring-license PIU sensitivity.

A.6 Spirit as the one-time package benchmark

Spirit remains the paper's most information-rich transaction because the public bankruptcy record describes the asset schedule, selected bid, alternate bid, deidentification obligation, and combined data/software package [18]. Google Cloud is used as an illustrative commercial surface because public segment economics are available. Under the 5% Google Cloud revenue-scope assumption, 35.6% segment operating-margin proxy, three-year life, 12% discount rate, $C = 0$, and $\rho = 1$, the selected \$10 million package bid implies PIU of approximately 0.236%. The 1%–10% scope sensitivity spans approximately 1.181% to 0.118%. This remains a package-level sensitivity because the filing does not allocate the bid between software and data and Google bears deidentification costs beyond the headline bid.

A.7 Scope of the transaction panel

The cases provide three kinds of external support. First, they establish observable AI-data/content consideration in the tens of millions of dollars per year. Reddit's securities filing additionally confirms a \$203 million aggregate licensing book with two- to three-year terms at the time Reuters reported the Google deal [36]. Wiley has separately disclosed recurring AI licensing revenue and individual AI projects, adding another public signal of a material market [45].

Second, Equation 2 can accommodate heterogeneous transaction structures while keeping assumptions visible. Recurring licenses yield a simple annual form when fee and benefit share the same timing. One-time acquisitions retain the discounted one-time formula. Caps produce upper-bound PIUs. Private loss-making buyers require a forward model.

Third, the panel supplies no controlled treatment/control estimate of Capability Alpha. A commercial license can reflect training value, retrieval/display value, legal certainty, product differentiation, strategic option value, or several channels at once. The supported empirical claim is narrower: PIU can be calculated from observed prices when buyer economics are available, yielding a break-even threshold for later data assays or production evaluation.

Comparability remains limited. The contracts bundle different rights, asset types, payment forms, and usage modes. Reported consideration may be approximate, and confidentiality prevents verification of many contractual details. The standardized 5% buyer-wide scope is intentionally mechanical. Consolidated operating margin is a proxy for cross-case comparison and does not represent the observed incremental margin of the affected AI product.

Market and financial sources for this appendix were accessed through August 25, 2026. Press-reported transaction values are used as reported market signals; the paper distinguishes them from contractual disclosures by the parties.

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